



GIK INSTITUTE

OF ENGINEERING SCIENCES AND TECHNOLOGY

Name: Abdullah Ejaz Janjua
Reg: 2023038
Instructor: Dr. Khurram Khan Jadoon

FACULTY OF COMPUTER SCIENCE AND ENGINEERING

Ghulam Ishaq Institute of Engineering Sciences and Technology, Topi, Swabi

Question 02

1. See the figure [1](#)
2. see function `value_iteration()` in `q2.py` file
3. see function `policy_iteration()` in `q2.py` file
4.
 - **Challenge: Value Initialization/Backup.** Ensuring the calculation for the next iteration uses the state values from the previous step ($V_k(s')$).
 - **MDP Concept:** State Value Function $V_k(s')$.
 - **Resolution:** Use of `current_values = copy.deepcopy(VALUE)` at the start of the outer iteration loop to guarantee a consistent backup of values.
 - **Challenge: Action-Dependent Rewards.** Correctly applying the driving cost (-1) only when the "Drive" action is taken.
 - **MDP Concept:** Action-Value Function $Q(s, a)$.
 - **Resolution:** Resetting `immediate_reward = REWARDS[state]` **inside** the `for action in ACTIONS` loop to ensure the cost is not incorrectly applied to other actions.
 - **Challenge: Policy Iteration Convergence.** Determining the point at which the optimal policy (π^*) has been found.
 - **MDP Concept:** Policy Stability.
 - **Resolution:** Implementing the boolean flag `policy_stable` (initialized to `True`) and setting it to `False` if the `best_action` found in the improvement step does not match the `old_action` ($\pi(s)$) for any state.

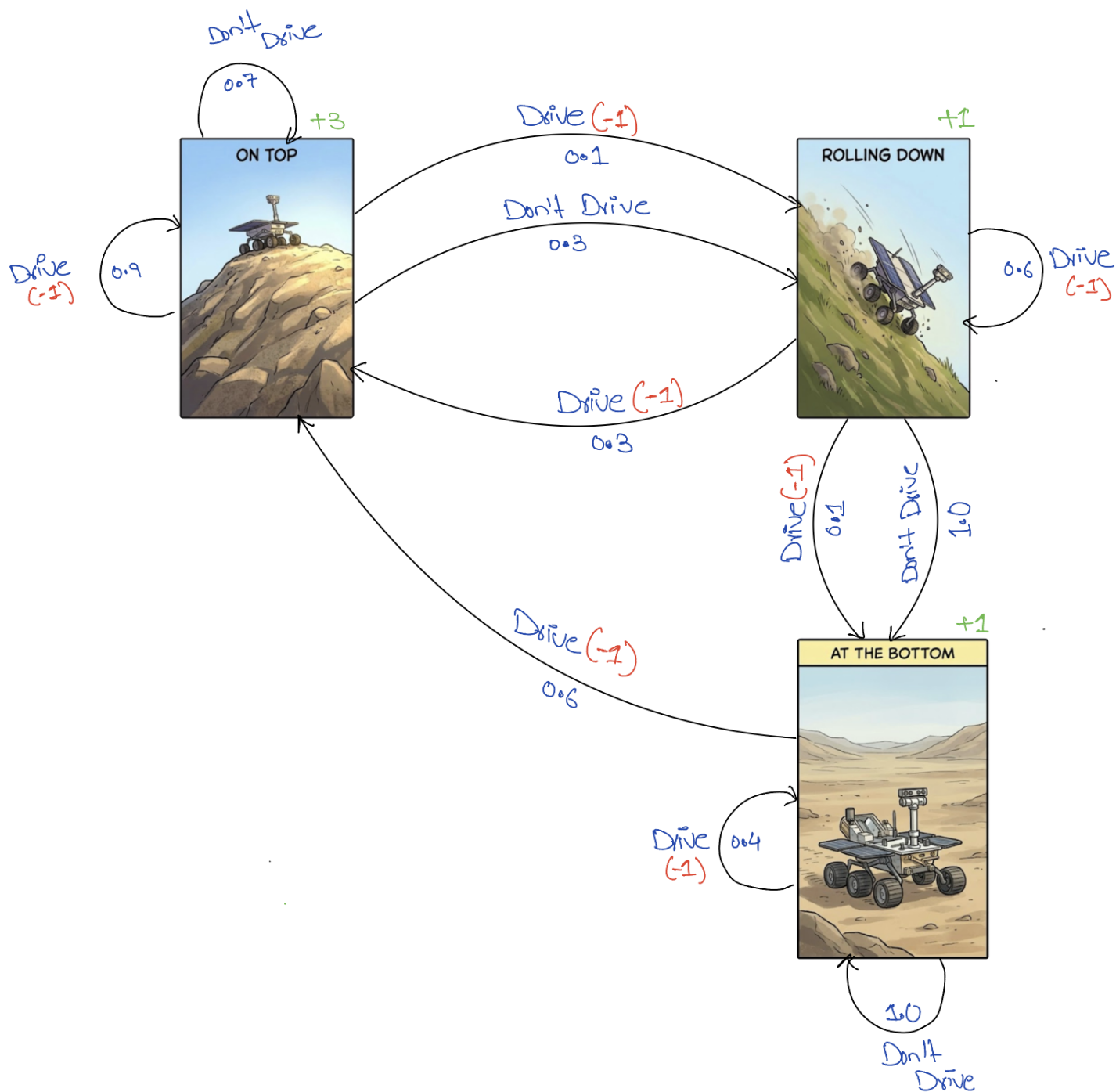


Figure 1: MDP represented graphically for the rover scenario in the Question